

FREDDY CASTRO

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EDUCATION

CUNY Graduate Center, New York, New York February 2024
Master of Science, Data Analysis and Visualization

SUNY Stony Brook University, Stony Brook, New York December 2017
Bachelor of Science, Applied Mathematics and Statistics (GPA: 3.38/4.00)

RESEARCH & WORK EXPERIENCE

STEM Instructor Brooklyn, NY
Kodely LLC August 2025 – Present

- Teaching students the fundamentals of coding using Minecraft Education Edition by introducing functions, sequencing, logic, and the coding agent in ways that strengthen computational thinking
- Guide students through weekly hands-on coding projects that improve problem solving skills and increase successful completion of programming tasks across the class by more than 80% over the course of the program
- Analyze class level learning trends to identify common areas of difficulty and share insights with the instructional team

Intergovernmental Affairs Liaison Intern
Queens, NY

Office of New York State Assemblymember Steven Raga Aug. 2025 – Dec. 2025

- Prepared analytical reports and summaries for legislative staff, including supplier comparisons, budget analyses, and cost-benefit reviews, to brief leadership
- Coordinated with intergovernmental partners to support data-driven policy and project planning
- Synthesized complex regulatory and statistical information into accessible presentations for diverse audiences
- Worked with cross-functional teams to transform data findings into actionable strategies that supported community initiatives, informed legislative decisions, and strengthened interagency collaboration

PROJECTS

NYC Air Quality and Public Health Analysis February 2024

Thesis & Capstone | CUNY Graduate Center | [GitHub](#)

- Conducted statistical and geospatial analysis using R, Python, and QGIS to evaluate public health trends
- Built a public-facing interactive website using HTML, CSS, JavaScript, and D3.js to reveal pollution inequities across NYC; findings informed community discussions on air quality and environmental justice
- Combined datasets from NYC Open Data, EPA, and CDC; created visualizations and maps that informed policy analysis; focused on environmental justice and data accessibility for community engagement

Asteroid Hazard Prediction with Machine Learning January 2024

Class Final Project | CUNY Graduate Center | [GitHub](#)

- Applied principal component analysis and trained classification models on NASA asteroid data to improve detection of hazardous asteroids; improved recall for asteroid detection while reducing false positives
- Applied logistic regression and decision trees; achieved accuracy of 99.98%
- Engineered features, handled missing data, and used PCA and clustering (k-means, DBSCAN, agglomerative)
- Practiced cross-validation, model evaluation, and dimensionality reduction for real-world predictive insights

2024

Exploratory and Narrative Data Stories | CUNY Graduate Center | [GitHub](#) & [Tableau](#)

- Designed responsive dashboards with D3.js, HTML, and CSS to visualize income disparity and urban patterns
- Used JavaScript to add interactivity and storytelling to data presentations hosted on GitHub Pages
- Emphasized clarity, accessibility, and visual narrative in communicating complex datasets

COMPUTER & TECHNICAL SKILLS

Programming: Python (Pandas, Scikit-Learn, Matplotlib, NumPy), R, JavaScript (D3.js), HTML, CSS

Statistical Modeling/Methods: Linear and Logistic Regression, Poisson Regression, ANOVA, Hypothesis Testing, Time Series Analysis, Decision Trees, PCA, Clustering (k-means, DBSCAN)

Tools and Platforms: Jupyter Notebooks, RStudio, QGIS, VS Code, Tableau, GitHub, Microsoft 365, Google Apps